

MoLAG: monocular LED assignment via a graph neural network for multi-tracker surgical navigation

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Abstract

Purpose: Monocular peer-to-peer tracking must group an unordered, scene-wide set of localised light-emitting diode (LED) points by source tracker before tracker-specific LED identification and point-to-line pose estimation. MoLAG applies a graph neural network to this grouping task.

Methods: MoLAG uses complete directed message passing and a symmetric affinity head to assign a same-tracker score to every unordered pair of detected points within a scene. Thresholded affinities form candidate tracker components. A scaled-conjunction loss promotes within-tracker connectivity, between-tracker separation, and suppression of spurious connections. Six full-fidelity loss settings were trained on generated coordinate scenes and evaluated on a frozen 50,000-scene synthetic test suite. A separate replicated hyperparameter search explored the loss family at reduced training fidelity.

Results: The combined full-fidelity setting achieved 89.64% complete-tracker extractability [89.35, 89.94] among 51,817 complete seven-LED trackers. Strict scene grouping was 56.43% [55.99, 56.86], all-tracker strict recovery was 76.20%, and real-real edge F_1 was 92.34%. Among the six full-fidelity settings, the combined setting produced the highest measured strict grouping and extractability. The reduced-fidelity search identified a replicated high-extractability operating region.

Conclusion: MoLAG effectively partitions multi-tracker LED detections on the evaluated synthetic coordinate distribution and retains most complete constellations for subsequent geometry-specific processing.

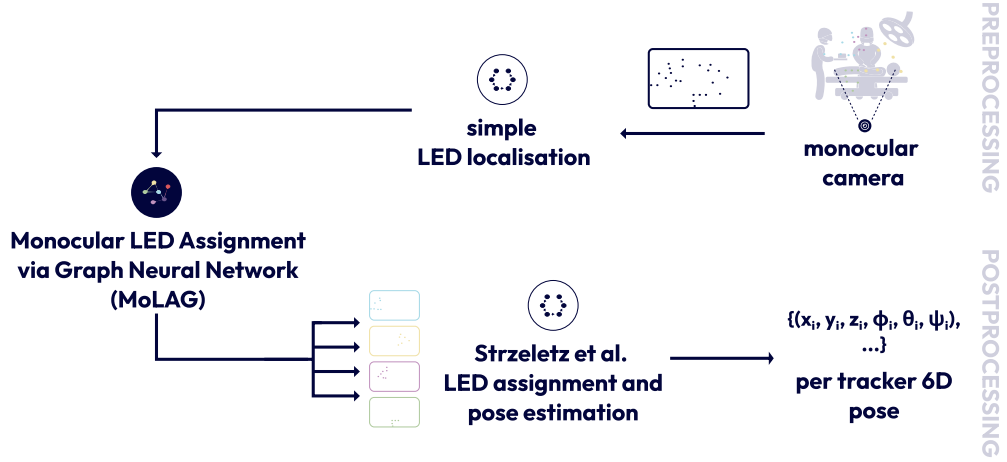


Fig. 1 MoLAG in the tracking pipeline. Preprocessing localises 2D blob centres. MoLAG partitions this scene-wide set into candidate tracker groups. The method of Strzeletz et al. [15] then identifies the LEDs and estimates each pose by point-to-line optimisation

Keywords: surgical navigation, optical tracking, LED tracker grouping, graph neural network, affinity learning, synthetic data

1 Introduction

Optical tracking is widely used in surgical navigation [3, 11]. Commercial systems observe rigid marker arrays attached to instruments and patient reference frames. Stereo systems triangulate the markers and register them to the known geometry, making pose availability dependent on marker visibility and array design [6, 20].

Strzeletz et al. [15] proposed a peer-to-peer architecture in which each device acts as both a tracked object and a tracking device, using a monocular camera to observe nearby devices. Each frame yields an unordered set of 2D detections from all visible trackers, possibly including spurious points. Tracker membership and LED identity are unknown. Their rule-based method forms marker-triplet hypotheses over the complete set. The number of hypotheses grows cubically with the number of detections. LED identification and pose estimation also require a complete seven-LED constellation. Crowding therefore enlarges the search, while missing detections reduce tracker availability.

MoLAG (**M**onocular **L**ED **A**ssignment via **G**raph neural network) partitions these detections by source tracker before tracker-specific LED identification and pose estimation (Fig. 1). Complete directed message passing provides scene context, a symmetric head predicts same-tracker affinities for every unordered detection pair, and thresholded connected components form the groups. Training uses generated coordinate sets with perspective variation, missing LEDs, spurious detections, and localisation noise rather than camera images.

The contributions are (1) a geometry-independent affinity architecture for variable-size multi-tracker point sets; (2) a scaled-conjunction loss that promotes within-tracker connectivity, tracker separation, and control of spurious connections; and (3) a full-fidelity loss study, replicated proxy-fidelity hyperparameter search, and detailed analysis of grouping performance on the evaluated synthetic distribution.

2 Related work

Monocular optical tracking reduces the camera-system footprint. Recent approaches use retroreflective fiducials [12], estimate instrument pose from images [9], or decode temporally modulated LEDs with an event camera [4]. These methods localise points, infer pose directly, or resolve LED identity through hardware signals. MoLAG instead groups an existing, unlabelled set of point detections.

The intended downstream method identifies geometric marker triplets and estimates pose by Lagrange-Newton point-to-line optimisation [14, 15]. Its square seven-LED marker differs from the triangular geometry evaluated here. Geometric LED correspondence also appears in monocular aerial tracking [5, 16]. These methods encode a specific marker layout, whereas MoLAG predicts tracker membership before geometry-specific processing.

MoLAG combines two established ideas. The Dynamic Graph CNN (DGCNN) introduced EdgeConv for message passing on point sets [19], and neural association methods predict pairwise relations before forming groups or instances [2, 18]. MoLAG uses a complete directed graph for contextual node embeddings and a symmetric affinity head for unordered detection pairs.

Correlation clustering and multicut methods enforce a globally consistent partition of pair relations [1, 13]. MoLAG instead uses direct thresholding and connected components for lightweight group extraction.

Independent affinity errors need not preserve the partition obtained after thresholding. Maximin affinity learning therefore trains graph connectivity rather than isolated edges [7, 17], while spanning-tree objectives retain within-class connectivity with few required edges [21]. The scaled-conjunction objective adopts these principles for tracker connectivity and separation, with additional terms for spurious attachments and paths.

3 Method

3.1 Task and evaluated tracker

For one monocular frame, MoLAG receives an unordered set of localised points

$$\mathcal{X} = \{\mathbf{x}_i\}_{i=1}^N, \quad \mathbf{x}_i \in \mathbb{R}^2, \quad (1)$$

where N is the number of detected points. The set contains LEDs from several trackers and possible spurious detections. MoLAG predicts a same-tracker affinity for every unordered pair of detected points and returns connected components as candidate groups. A downstream method can then identify LEDs and estimate pose within each

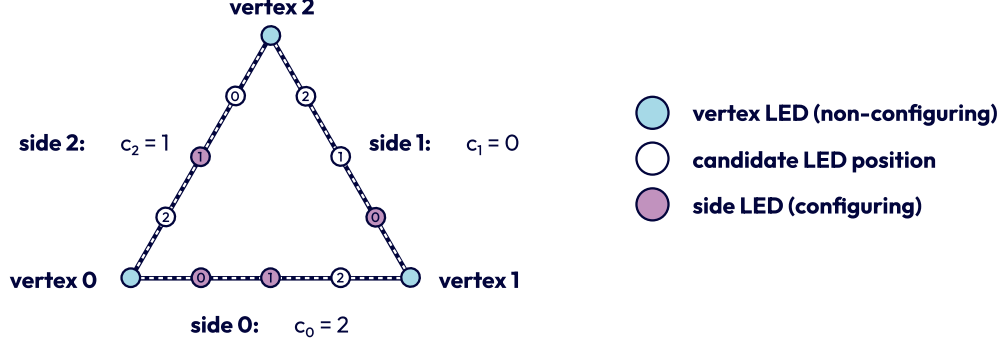


Fig. 2 Evaluated seven-LED tracker for code $\mathbf{c} = (2, 0, 1)$. Three fixed vertex LEDs and four code-dependent side LEDs form one of 27 configurations

group. The grouping formulation depends only on coordinates and tracker-membership labels, not a marker template.

The experiments use a triangular seven-LED tracker with a 64-mm side length (Fig. 2). Three LEDs occupy the vertices. A code $\mathbf{c} = (c_0, c_1, c_2)$ selects four side LEDs from three candidate positions per side: c_1 and c_2 select one position each, while c_0 identifies which of the three candidate positions on side 0 is omitted, leaving the other two occupied. This asymmetric encoding gives $3^3 = 27$ configurations. Other layouts require representative training data but no architectural change.

3.2 Coordinate-level scene generation

Training scenes are generated at MoLAG’s image-point interface. Rigid 3D trackers are sampled, projected through a monocular pinhole camera, and modified by LED dropout, spurious points, and Gaussian coordinate noise. Inter-tracker body occlusion is rejected geometrically. Each point set is mean-centred and divided by its maximum radius, $\tilde{\mathbf{x}}_i = (\mathbf{x}_i - \bar{\mathbf{x}}) / \max_j \|\mathbf{x}_j - \bar{\mathbf{x}}\|_2$. This removes translation and uniform scale while preserving relative geometry, where $\bar{\mathbf{x}}$ is the mean of $\mathcal{X}$. The exact camera, pose, and error distributions are reported in Online Resource 1 (Supplementary Information), Sect. S2.2.

Fig. 3 summarises the generation sequence.

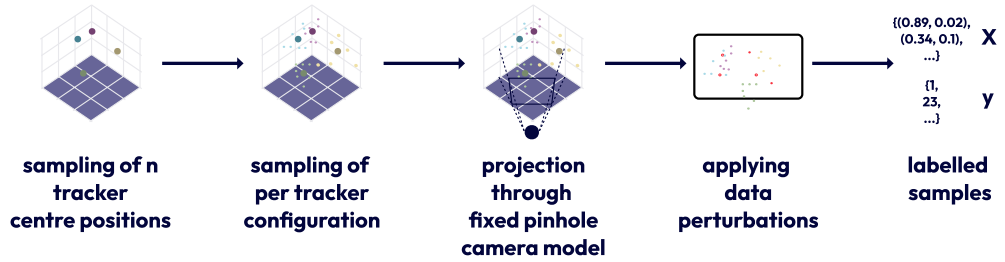


Fig. 3 Coordinate-level scene generation. Triangular trackers are placed at sampled poses, checked for body occlusion, and projected through the camera model. LED dropout, spurious points, normalisation, and Gaussian noise then model the point-localisation interface

3.3 Affinity graph network

Two edge sets distinguish message passing from affinity prediction. The first contains both directions between every pair of detections; the second contains each unordered pair once:

$$\begin{aligned}\mathcal{V} &= \{1, \dots, N\}, \\ \mathcal{E}_{\text{mp}} &= \{(j \rightarrow i) : i, j \in \mathcal{V}, i \neq j\}, \\ \mathcal{E}_{\text{aff}} &= \{\{i, j\} : i, j \in \mathcal{V}, i < j\}.\end{aligned}\tag{2}$$

These sets define the complete directed message-passing graph $\mathcal{G}_{\text{mp}} = (\mathcal{V}, \mathcal{E}_{\text{mp}})$ and the complete undirected candidate graph $\mathcal{G}_{\text{aff}} = (\mathcal{V}, \mathcal{E}_{\text{aff}})$. Thus, every point receives information from every other point, whereas one affinity is predicted per unordered pair.

Each node starts with its normalised coordinate, $\mathbf{h}_i^{(0)} = \tilde{\mathbf{x}}_i$. A directed edge $i \rightarrow j$ carries squared distance and direction:

$$\mathbf{e}_{i \rightarrow j} = \left[\|\tilde{\mathbf{x}}_j - \tilde{\mathbf{x}}_i\|_2^2 \parallel \frac{\tilde{\mathbf{x}}_j - \tilde{\mathbf{x}}_i}{\max(\|\tilde{\mathbf{x}}_j - \tilde{\mathbf{x}}_i\|_2, \varepsilon_{\text{num}})} \right]. \tag{3}$$

$\varepsilon_{\text{num}} > 0$ prevents division by zero for coincident points. Seven EdgeConv-inspired message-passing blocks [8, 19] update every node by component-wise maximum aggregation over its incoming messages:

$$\mathbf{h}_i^{(\ell+1)} = \max_{j \neq i} \phi_{\theta_{\text{mp}}^{(\ell)}} \left([\mathbf{h}_i^{(\ell)} \parallel \mathbf{h}_j^{(\ell)} \parallel \mathbf{e}_{j \rightarrow i}] \right). \tag{4}$$

Here $\ell = 0, \dots, 6$ indexes the blocks, and each $\phi_{\theta_{\text{mp}}^{(\ell)}}$ is a two-layer ReLU network with parameters $\theta_{\text{mp}}^{(\ell)}$. The block widths are 128, 256, 512, 1,024, 2,048, 2,048, and 1,024. Concatenating all block outputs gives

$$\mathbf{h}_i = [\mathbf{h}_i^{(1)} \parallel \dots \parallel \mathbf{h}_i^{(7)}] \in \mathbb{R}^{7,040}. \tag{5}$$

For each $\{i, j\} \in \mathcal{E}_{\text{aff}}$, a two-layer multilayer perceptron with 128 hidden units and an intermediate ReLU evaluates the symmetric representation

$$z_{ij} = f_{\theta}([\mathbf{h}_i \odot \mathbf{h}_j \parallel |\mathbf{h}_i - \mathbf{h}_j|]), \quad p_{ij} = \sigma(z_{ij}), \tag{6}$$

Here, $\odot$ and $\parallel$ denote the elementwise product and concatenation; f_{θ} is the affinity MLP with parameters θ ; and σ is the logistic sigmoid. The representation is invariant to exchanging the two points. The predictions weight the complete affinity graph $\hat{\mathcal{G}}_{\text{aff}} = (\mathcal{V}, \mathcal{E}_{\text{aff}}, p)$. Fig. 4 summarises the architecture. Online Resource 1, Sect. S2.1 provides the corresponding implementation and training details.

3.4 Scaled-conjunction affinity objective

For scene s , $V_{s,t}$ contains the visible nodes of tracker t , $\mathcal{T}_s$ the tracker set, and B_s the spurious nodes. Independent pair supervision does not directly encode the required

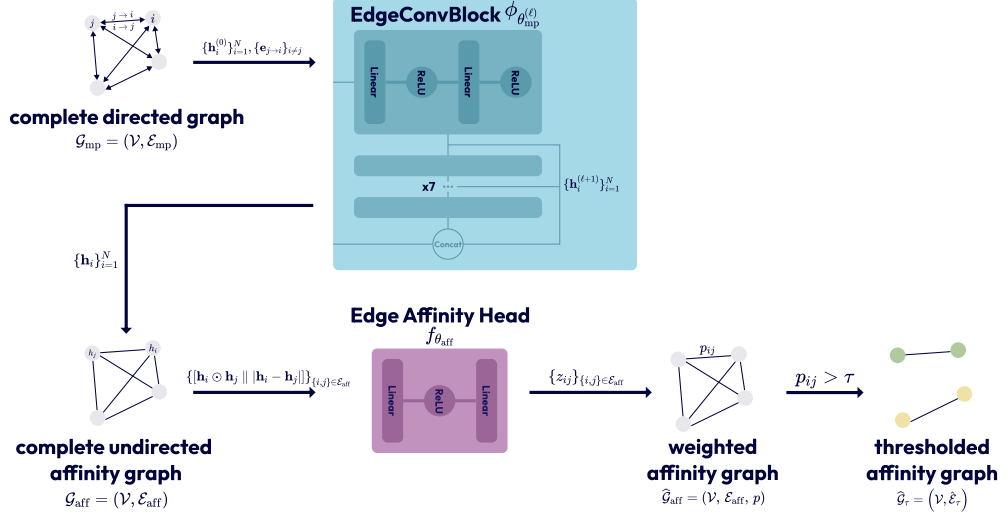


Fig. 4 MoLAG architecture. Complete directed message passing forms node embeddings. A symmetric head predicts affinities for unordered pairs, and connected components of the thresholded graph define candidate groups

partition: each tracker needs only enough positive edges to remain connected, while one cross-tracker edge can merge two components. The structured objective therefore combines tracker connectivity, tracker-pair separation, spurious attachment, and spurious-bridge conditions.

Within a condition, edge losses v_a with non-negative handicaps η_a , indexed by $a \in A$, are reduced by the weighted soft maximum

$$\mathcal{M}_\beta(v, \eta; A) = \frac{1}{\beta} \left[\log \sum_{a \in A} e^{\beta(v_a - \eta_a)} - \log \sum_{a \in A} e^{-\beta \eta_a} \right]. \quad (7)$$

Finite β distributes gradient across poor edges; $\beta \rightarrow \infty$ recovers the worst handicapped edge. For a mini-batch of S scenes, let n_s be the number of applicable conditions in scene s , with losses $q_{s,\kappa}$. The scene reduction is

$$\mathcal{R}_{\alpha,\rho}(q) = \frac{\sum_{s \in \mathcal{S}^*} n_s^\alpha \left(n_s^{-1} \sum_{\kappa=1}^{n_s} q_{s,\kappa} \right)}{(1-\rho)S + \rho|\mathcal{S}^*|}, \quad \mathcal{S}^* = \{s : n_s > 0\}. \quad (8)$$

The power $\alpha \in [0, 1]$ controls the influence of crowded scenes, and ρ determines whether scenes without an applicable condition enter the outer mean.

The connectivity term uses a maximum-affinity spanning tree for each tracker. The separation term penalises the strongest edge between each tracker pair. The attachment term penalises the strongest link from a spurious point to a tracker, and the bridge term penalises the widest path between trackers through spurious nodes. An

optional scene-local supervised contrastive (SupCon) term structures real-node embeddings [10]. With non-negative term weights w_c , w_s , w_b , and w_{sc} , and attachment scale $\varepsilon_{\text{spur}}$, the complete loss is

$$\mathcal{L} = w_c \mathcal{L}_{\text{conn}} + w_s (\mathcal{L}_{\text{pair}} + \varepsilon_{\text{spur}} \mathcal{L}_{\text{att}}) + w_b \mathcal{L}_{\text{bridge}} + w_{sc} \mathcal{L}_{\text{supcon}}. \quad (9)$$

Online Resource 1, Sect. S3 defines every loss term and boundary case. It also defines the connectivity, separation, and bridge margins m_c , m_s , and m_b ; the non-tree discount δ ; the SupCon temperature τ_{sc} ; and the condition-specific powers α_c for connectivity and bridges and α_s for separation. The combined full-fidelity setting uses $w_c = 1$, $w_s = 0.46$, $w_b = 0.25$, $m_c = m_s = 1$, $m_b = 0$, $\beta = 1$, $\delta = 3$, $\varepsilon_{\text{spur}} = 0.01$, $\alpha_c = \alpha_s = 0.5$, $\rho = 1$, and $w_{sc} = 0$.

3.5 Group extraction

At inference, a threshold selected on held-out calibration data defines

$$\hat{\mathcal{E}}_\tau = \{\{i, j\} \in \mathcal{E}_{\text{aff}} : p_{ij} \geq \tau\}, \quad \hat{\mathcal{G}}_\tau = (\mathcal{V}, \hat{\mathcal{E}}_\tau). \quad (10)$$

The connected components of $\hat{\mathcal{G}}_\tau$ form the candidate groups; isolated points form singleton components. The resulting partition is deterministic given the network predictions.

4 Experiments

4.1 Training and evaluation protocol

The study evaluates overall grouping, six full-fidelity loss configurations, a proxy-fidelity hyperparameter optimisation (HPO) search, and performance across scene compositions. Each full-fidelity run used the same architecture and 5,000,000 generated scenes for ten epochs. Scenes contained one to ten distinct tracker configurations. The generator sampled tracker centres within $x, y \in [-200, 200]$ mm and $z \in [100, 600]$ mm, face tilts up to 85° , LED dropout, up to four spurious detections, and coordinate noise. All generation and training parameters are reported in Online Resource 1, Sects. S2.1 and S2.2.

Each checkpoint was calibrated on 5,000 frozen scenes, with 500 for every tracker count from one to ten. Thresholds from 0.05 to 0.95 were evaluated in steps of 0.01, and the threshold with the highest strict scene grouping was retained. Final evaluation used a disjoint frozen suite of 50,000 scenes, stratified into 5,000 scenes for each exact tracker count. Training, in-training evaluation, calibration, HPO validation, and final test scenes were mutually disjoint. All checkpoints used the same test scenes, enabling paired comparisons.

4.2 Evaluation metrics

Let $\hat{g}_s(i)$ denote the predicted component of node i in scene s . Tracker t is strictly recovered when all nodes in $V_{s,t}$ share one component and that component contains

no real node from another tracker. Let $r_{s,t}$ indicate this event. For M_{test} test scenes, strict scene grouping and all-tracker strict recovery are

$$A_{\text{group}} = \frac{1}{M_{\text{test}}} \sum_s \prod_{t \in \mathcal{T}_s} r_{s,t}, \quad R_{\text{all}}^{\text{strict}} = \frac{\sum_s \sum_{t \in \mathcal{T}_s} r_{s,t}}{\sum_s |\mathcal{T}_s|}. \quad (11)$$

All sums in these metric definitions are over the test scenes. A scene therefore succeeds only if every tracker is unsplit and unmerged. Spurious detections may attach to one tracker but cause failure if they connect trackers.

Complete trackers contain all seven LEDs. Let $\mathcal{T}_s^{(7)} = \{t \in \mathcal{T}_s : |V_{s,t}| = 7\}$ denote the complete trackers in scene s . Complete-tracker strict recovery applies the same criterion to this subset. Complete-tracker extractability is the number of distinct predicted components containing at least one unsplit complete tracker, divided by the number of complete trackers. For an unsplit tracker, let $\hat{g}_{s,t}$ denote its component:

$$E_{\text{complete}} = \frac{\sum_s \left| \left\{ \hat{g}_{s,t} : t \in \mathcal{T}_s^{(7)}, |\{\hat{g}_s(i) : i \in V_{s,t}\}| = 1 \right\} \right|}{\sum_s |\mathcal{T}_s^{(7)}|}. \quad (12)$$

A split tracker contributes zero; a component containing two complete trackers contributes once. Extractability is thus a merge-tolerant grouping proxy, conditional on a downstream method being able to identify an intact constellation among additional detections.

Conventional precision, recall, and F_1 quantify pairwise affinities. Real-only grouping removes edges incident to spurious nodes before component extraction. Further diagnostics distinguish splits, direct merges, mixed errors, spurious-mediated bridges, and spurious attachments. Online Resource 1, Sect. S4 defines these diagnostics and gives illustrative cases.

4.3 Loss-configuration studies

The full-fidelity study trained one conjunction reference and five variants. All runs shared the network, data, optimiser, checkpoint rule, calibration, and test suite; Table 1 lists every changed loss parameter. Each of the first four variants addresses one design question relative to the reference. The combined scaled-conjunction configuration changes several parameters and is interpreted only as a complete setting. Across all rows, $w_c = 1$, $m_c = m_s = 1$, $w_b = 0.25$, $m_b = 0$, $\tau_{\text{sc}} = 0.2$, and the maximum tracker size is seven LEDs. Online Resource 1, Sect. S3 defines these fixed parameters and the parameters in Table 1.

The exploratory search examined 77 settings at 6 % training fidelity. Three retained candidates and the combined setting as proxy incumbent were then trained three times each. Threshold selection and scoring used disjoint halves of a frozen 20,000-scene suite, subject to a strict-grouping floor of 48.39 %. This study measures replicated proxy-fidelity behaviour, not transfer to full-fidelity training. Its complete design and record appear in Online Resource 1, Sect. S7.

Table 1 Changed loss parameters in the full-fidelity study. The loss definitions and fixed parameters are given in Online Resource 1, Sect. S3

Configuration	β	δ	α_c	α_s	ρ	w_s	w_{sc}	$\varepsilon_{\text{spur}}$
Conjunction reference	∞	∞	0	0	0	2.00	0.03	0
Dense gradient	1	3	0	0	0	2.00	0.03	0
Reduced separation	∞	∞	0	0	0	0.90	0.03	0
Scene weighting	∞	∞	0.5	0.5	1	1.00	0.03	0
No SupCon	∞	∞	0	0	0	2.00	0	0
Combined scaled conjunction	1	3	0.5	0.5	1	0.46	0	0.01

Table 2 Combined-configuration performance on the frozen test suite. Brackets contain 95 % intervals

Measure	Result
Strict scene grouping	56.43 % [55.99, 56.86]
Real-only grouping	56.33 %
All-tracker strict recovery	76.20 %
Complete-tracker strict recovery	80.36 % [79.89, 80.83]
Complete-tracker extractability	89.64 % [89.35, 89.94]
All-pair edge F_1	90.84 %
Real-real edge precision / recall / F_1	94.33 % / 90.43 % / 92.34 %

4.4 Statistical analysis

Strict scene grouping uses 95 % Wilson intervals. Complete-tracker ratios use scene-clustered delta-method intervals because tracker outcomes within a scene are correlated. Paired McNemar tests compare full-fidelity checkpoints on common binary scene outcomes; these single-run comparisons do not estimate training variation. The proxy study reports means and sample standard deviations across three runs per arm. The primary selected-candidate versus proxy-incumbent contrast uses pooled within-arm variation. Online Resource 1, Sects. S4 and S7.3 report the complete procedures and statistics.

5 Results

5.1 Overall performance

The test suite contained 51,817 complete trackers among 275,000 tracker instances. The combined full-fidelity configuration retained 89.64 % as extractable constellations. Complete-tracker strict recovery was 80.36 %, all-tracker strict recovery was 76.20 %, and 56.43 % of the 50,000 scenes were grouped exactly (Table 2). The difference between extractability and strict recovery shows that intact constellations often remained available inside imperfect partitions.

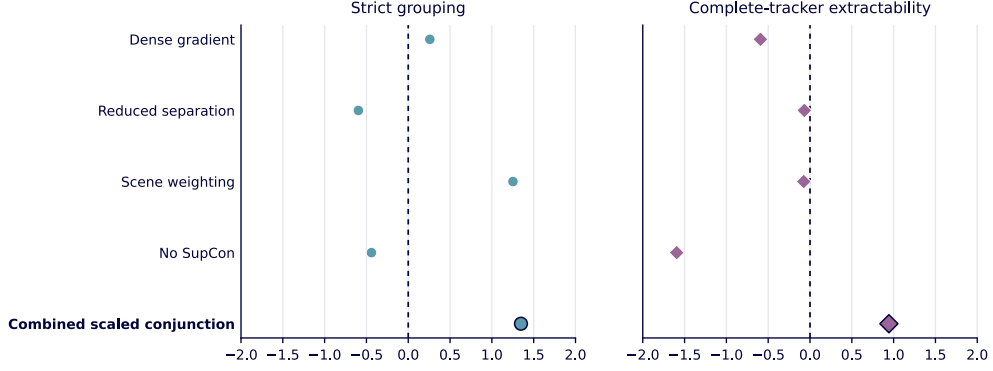


Fig. 5 Percentage-point differences from the conjunction reference in strict scene grouping and complete-tracker extractability. Each point is one checkpoint

5.2 Loss configurations

The combined checkpoint led both primary outcomes among the six full-fidelity settings (Fig. 5). Among the first four variants, scene weighting showed the largest increase in strict grouping while preserving extractability. Dense gradient sharing and reduced separation shifted the balance between the outcomes, whereas removing SupCon from the reference reduced both. Each setting was trained once, so these results compare checkpoints rather than training distributions; the combined setting does not isolate individual parameter effects. Online Resource 1, Tables S1 and S2 report absolute results and paired statistics.

The combined setting subsequently served as the incumbent in the proxy-fidelity search. All three retained candidates exceeded its three-run mean extractability. In the primary comparison, which also retained an earlier incumbent run, the selected candidate exceeded the incumbent by 2.14 percentage points [1.21, 3.07] after accounting for measured between-run variation at this fidelity. Strict grouping and all-tracker strict recovery were similar, while edge F_1 decreased and spurious attachment increased. No search-selected configuration was trained at full fidelity. Online Resource 1, Sect. S7 reports the complete search and replication.

5.3 Scene composition and failures

Crowding affected exact grouping more strongly than extractability (Fig. 6). Strict grouping fell from 97.58 % with one tracker to 12.32 % with ten, whereas complete-tracker extractability remained 84.46 % in the ten-tracker stratum. Merges and mixed split-and-merge errors dominated failed scenes; isolated splits and failures caused only by spurious bridges were rare. Visibility, spurious detections, tracker configuration, total point count, and the full failure composition are analysed in Online Resource 1, Sect. S5.2.

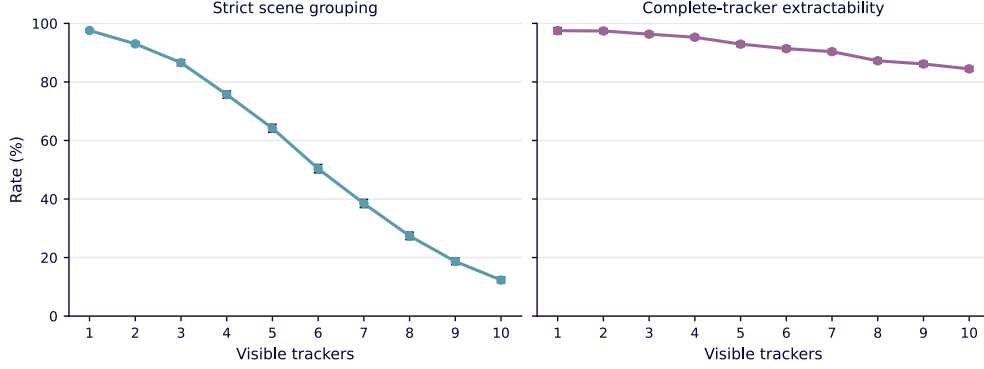


Fig. 6 Strict scene grouping and complete-tracker extractability by exact visible tracker count. Each stratum contains 5,000 scenes

6 Discussion

MoLAG partitions scene-wide detections before tracker-specific LED identification and pose estimation. Its 89.64% extractability among 51,817 complete trackers is the central result for this role: most complete constellations remained intact in one candidate component, including in crowded scenes. Strict grouping was lower because it requires every tracker in a scene to be both unsplit and unmerged. The widening gap between these measures with tracker count identifies separation, rather than loss of complete constellations, as the main residual challenge.

The full-fidelity comparison supports the combined configuration as the strongest of the six evaluated checkpoints, but does not isolate the effects of its changed parameters. Independent repeats would be required for configuration-level claims. The replicated proxy-fidelity search identifies a stable high-extractability region, although lower edge precision and greater spurious attachment show the associated operating-point trade-off. These settings have not been tested at full fidelity.

The evidence applies to generated coordinate sets, not recorded images. The generator varies perspective, tracker count, dropout, clutter, and localisation noise while preserving exact membership labels. Its random dropout, uniform spurious points, Gaussian noise, ideal pinhole camera, and distinct configuration codes do not reproduce the spatial and appearance structure of an operating room. Translation and scale normalisation also remove global image position and apparent size. Transfer to real localisation output, repeated codes, and other tracker geometries therefore remains to be established.

The study evaluates independent frames and grouping outcomes. It does not measure temporal association, cross-camera fusion, runtime, LED identification, or pose accuracy. Extractability further assumes that the downstream geometric method can identify a complete constellation within a component containing additional detections; an equivalent procedure for the triangular tracker remains to be integrated. The number of predicted affinities grows quadratically with detected points, so the net computational effect depends on the downstream implementation. The single-run

full-fidelity comparison and reused HPO selection split also leave training variation and selection optimism unresolved.

An end-to-end study should connect recorded LED localisation, MoLAG, tracker-specific identification, and point-to-line pose estimation. Measuring pose availability and accuracy, runtime, and localisation-error distributions will test the intended search reduction and support calibration of the synthetic training distribution.

7 Conclusion

MoLAG combines complete message passing, symmetric affinity prediction, and a scaled-conjunction loss to partition multi-tracker LED detections. On 50,000 generated test scenes, it retained 89.64 % of 51,817 complete trackers as extractable constellations, with 76.20 % all-tracker strict recovery and 56.43 % exact scene grouping. These results establish effective grouping on the evaluated coordinate distribution. The next step is end-to-end evaluation with recorded localisation output and downstream LED identification and pose estimation.

Statements and Declarations

- **Funding:** No funding was received for conducting this study.
- **Competing interests:** Yannik Heizmann and Harald Hoppe declare that they have no relevant financial or non-financial interests to disclose.
- **Ethics approval:** This article does not contain any studies with human participants or animals performed by either author.
- **Informed consent:** Not applicable because the study involved no human participants.
- **Data availability:** No patient or recorded camera data were used. All experimental input data were generated procedurally. Trained model weights and supporting benchmark and HPO artefacts are available on reasonable request.
- **Code availability:** The MoLAG implementation, generation methodology, documentation, and experiment configurations are available at <https://github.com/yannikheizmann/MoLAG/tree/v0.1.0> (version 0.1.0).
- **Author contributions:** Yannik Heizmann conceived the study, developed the method and software, designed and conducted the experiments, analysed and visualised the results, and wrote and revised the manuscript. Harald Hoppe contributed the initial project concept and reviewed the manuscript.
- **Use of generative AI:** OpenAI Codex was used to assist with language refinement, manuscript organisation, and consistency checks. The authors reviewed and verified the final manuscript and retain full responsibility for its content.

Supplementary Information. Online Resource 1: Complete architecture, loss and metric definitions, reproducibility settings, detailed results, diagnostic analyses, and hyperparameter optimisation record.

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